

Resources for Professional Development

Scholarship opportunities

1. ICTP STEP Program:
<https://www.ictp.it/opportunity/ictpiaea-sandwich-training-education-programme> (Funds PhD students from developing countries for visits). See also <https://www.ictp.it/math> for other math programs.
2. IMU-CDC <https://www.mathunion.org/cdc> (various types of funding opportunities and conferences)
 - a. IMU Breakout Graduate Fellowship Program
<https://www.mathunion.org/cdc/scholarships/graduate-scholarships/imu-breakout-graduate-fellowship-program>
 - b. GRAID
<https://www.mathunion.org/cdc/scholarships/graduate-scholarships/graduate-assistantships-developing-countries>
3. CIMPA <https://www.cimpa.info/> (schools, conferences, programs, etc.)
4. CoRE-Math <https://www.africaeuropecoremath.org/projects> (fellowships, schools)
5. LMS <https://www.lms.ac.uk/grants/international-grants> (grants)
6. EMS-CDC <https://euromathsoc.org/committee-developing-countries> (travel grants)
7. AIMS:
 - a. Next Einstein policy: <https://covid19.nexteinstein.org/about/>
 - b. Quantum Leap: <https://aims-network.org/programs/discovery/quantumleap/> (lists AIMS programs)
 - c. Fields-AIMS-Perimeter Africa Fellowship
<https://perimeterinstitute.ca/fields-aims-perimeter-africa-fellowship> (visiting programs and fellowships, primarily for PhD holders, but other things available too).
8. The World Academy of Science <https://twas.org> (programs, scholarships)
9. East Africa Institute for Fundamental Research <https://eaifr.ictp.it> (information about masters programs, scholarships)
10. International Science Programme <https://www.isp.uu.se> (international collaborations, funding programs)
11. KAUST postdoctoral fellowship: <https://kgfp.kaust.edu.sa/> (more focused on global challenges and social problems).
12. Funding for a short visit inviting researchers from “widening countries”
<https://www.ncpwideranet.eu/study-visits/> (only African countries included are Morocco and Tunisia)
13. Aga Khan Foundation Scholarship Program
(<https://akf.org/international-scholarship-programme/>)

14. Coimbra Scholarships for Young professors and Researchers:
<https://www.coimbra-group.eu/scholarships/>
15. Google DeepMind (<https://deepmind.google/education/>) - started a scholarship program for masters students joint with AIMS, specifically focused on AI. There are also some postgraduate scholarships
16. AI4D-IRDC program (<https://idrc-crdi.ca/en/initiative/artificial-intelligence-development>): AI focused grants. The IRDC also has other funding programs to keep an eye out on.

There are double or joint doctorates between African and (e.g.) European universities, where you spend time at both universities and often can get two PhD diplomas. The paperwork between universities can be very difficult, but if they and the potential supervisors can agree, this can sometimes be a simpler way to do a PhD (partially) abroad than going through the regular graduate program of the non-African university.

Programs focused on working in Germany:

1. DFG (German Research Foundation):
https://www.dfg.de/en/research_funding/programmes/international_cooperation/developing_countries/
 - a. Initiation of collaborations
https://www.dfg.de/en/research_funding/programmes/international_cooperation/initiation_international_collaboration/
2. DAAD (German Academic Exchange Service) in East Africa
<https://www.daad.de/en/the-daad/what-we-do/sustainable-development/funding-programmes/funding-programmes-for-students-a-z/scholarship-east-africa/> (might not exactly be relevant for mathematics, but might work well for the future, and for people interested in programs for social good)
3. AGNES (African-German Network of Excellence in Science)
<https://agnes-h.org/category/opportunities/> (funding opportunities, seems to have extra grants for women).
4. Max Planck Society <https://www.mpg.de/16384579/africa> (mentorship program)
5. Alexander von Humboldt Foundation
<https://www.humboldt-foundation.de/bewerben/alumnifoerderung/alumni-im-ausland/humboldt-forschungshubs-in-afrika> (some residency programs, funding decreased recently).
6. BMZ (Germany ministry for Development), currently all funding offers are closed, but worth keeping in mind <https://www.bmz.de/en>

Women focused

1. Organization for Women in Science for the Developing World (OWSD): <https://owsd.net/> (Good for: networking, events, prizes and scholarships)
2. African Women in Algebra (AWA): <https://sites.google.com/view/africanwomeninalgebra/home> (workshops, research groups, network, information).
3. <https://femafricmaths.github.io/>
4. Women in Sage: <https://wiki.sagemath.org/Workshops> (For women, but part of larger workshop series for everyone. Good for: workshops, research groups, network)
5. African Women in Mathematics Association (<https://www.africanwomeninmath.org/>) (Good for: workshops, network)
6. EWM Travel grant: <https://www.europeanwomeninmaths.org/travel-grants/> (Grants for women from developing countries to present at a conference)
7. L'Oreal UNESCO Women in Science Programme and Prize (<https://www.unesco.org/en/prizes/women-science>) (Prize and other regional programs)
 - a. See also: <https://scholar.africa/scholarships/loreal-unesco-women-in-science-sub-saharan-africa>
8. Seed grant New African Principal Investigator (contains Scientist after Child scheme) <https://twas.org/opportunity/seed-grant-new-african-principal-investigators-sg-napi>
9. May12 <https://may12.womeninmaths.org> Organize an event for International Women in Maths Day! Many ideas and resources in the Website!

Some networks based in the US/Canada/Europe

These can be helpful for building an international network and

1. Women in Number Theory <https://womeninnumbertheory.org> (Useful mailing list to join, collaborative workshops)
2. Women in Algebraic Geometry <https://awm-math.org/research-networks/wiag/> (collaborative workshops)
3. Association for Women in Mathematics <https://awm-math.org/> (US-focused, but sometimes useful for workshops, finding mentors, etc)
4. Women in Data Science <https://wids.nl> (Based in the Netherlands)
5. EDGE for Women in Maths <https://www.edgeforwomen.org/> (Fellowships and events, US focused)
6. STEM Sisters <https://www.linkedin.com/company/stem-sisters/> <https://istemco.com/>

Social impact jobs and opportunities

1. IDEAM International <https://www.idems.international>
2. INNODEMS <https://innodems.org> (job opportunities with high social impact)
3. African Maths Initiative <https://www.africanmathsinitiative.net> (NGO focused on math education)

Places that run regular mathematics programmes

..and that might have visiting positions. These are just a small sample, there are many more. Keep in mind that getting visas might be cumbersome.

1. CIRM <https://www.cirm-math.com>
2. MSRI <https://www.msri.org/web/cms>
3. ICMS Edinburgh: <https://icms.ac.uk/> (also have funding for workshops)
4. Banff International Research Station <https://www.birs.ca/programs> (they have multiple branches across the world - keep an eye on all locations)

Other

Inspiring stories

1. Youtube channel of Femafrika Maths.
2. EvenQuads: playing cards with women in science (<https://awm-math.org/publications/playing-cards/>) (Note: Submit names for more African women!)
3. Black Heroes of Mathematics (annual conference) <https://www.lms.ac.uk/events/black-heroes-of-mathematics-conference-2026>
4. Mathematically gifted and black <https://mathematicallygiftedandblack.com>

List of open access journals:

- <https://open-access.network/en/information/subject-specific-open-access/mathematics>

Conferences and schools in number theory/algebraic geometry

Keep in mind that not all conferences provide funding, and sometimes have restrictions on the number of participants from outside the country. Also, make sure you are interested in the topic of the conference that you apply for - organizers receive a lot of applications!

1. Rethinking number theory (online collaborative workshop on number theory): <https://sites.google.com/view/rethinkingnumbertheory/home>

2. Arizona Winter School (<https://swc-math.github.io/>) annual school in number theory in the US. They also have a lot of resources on their website, and they have an online program.
3. List of conferences in arithmetic geometry (<https://kskedlaya.org/cgi-bin/confs.cgi>). This also has several Cryptography conferences.
4. Conferences in algebraic geometry (<https://math.stanford.edu/~vakil/conferences.html>)

Mailing lists and websites for academic positions (focused on the US, UK and Europe)

Where the advertisements are, when the deadlines are, and how to apply varies from country to country. Here are some places

- <https://www.jobs.ac.uk/> (UK, all levels)
- US PhD positions: <https://www.phdportal.com/countries/82/united-states.html>
 - Deadlines are usually in early December
 - Check the graduate programs on each university website.
 - The US system is more flexible if you don't have a certain area or topic in mind, and PhDs are funded once admitted. But applying is expensive (GRE, TOEFL, etc. cost money), although some are now waiving this: <https://worldstudyhub.com/universities-in-usa-that-do-not-require-gre-gmat/>
 - Keep in mind that with the political situation in the US, visas are getting harder.
- To find postdocs in mathematics: <https://www.mathjobs.org/> (lots of countries)
- (General) <https://academicpositions.com/>
- To receive advertisements for PhD positions in Algebraic Geometry or Number theory in Europe you should subscribe to the following mailing lists:
 - The European algebraic geometry mailing list: <https://www.listserv.dfn.de/sympa/info/eager-gen>
 - The UK algebraic geometry mailing list: <http://listserv.csv.warwick.ac.uk/mailman/listinfo/cowalggeom>
 - The number theory mailing list: <https://listserv.nodak.edu/cgi-bin/wa.exe?A0=NMBRTHRY>

Not all positions are advertised through these mailing lists.

- Some doctoral schools:
 - Berlin Mathematical School <https://www.math-berlin.de/about-bms>
 - London Graduate School in Geometry and Number Theory <http://www.lsgnt-cdt.ac.uk/>
 - ISTA Graduate School <https://phd.pages.ist.ac.at/phd-program-overview/>
 - Glasgow Graduate School in Algebraic Structures <https://www.glams.org/home>

Some useful advice from [Ravi Vakil](https://math.stanford.edu/~vakil/potentialstudents.html): <https://math.stanford.edu/~vakil/potentialstudents.html>

CVs, Cover letters, etc.

In general, to apply for a PhD position you will be asked for a CV, a motivation letter, transcripts of your bachelor and master grades, and names of people who write your letters of references. You may need from 2 to 4 letters depending on where you apply. Usually your references would be from professors you have been working with closely, such as your master thesis advisor, bachelor thesis advisor, supervisor of some other project you worked on, teacher of one of your courses, tutor, etc. You have to ask these people whether they can write a letter for you, and you want to ask the people you think are going to write a good recommendation. You can also ask lecturers at a school like this one, but make sure that you interact with them, so that they know you better and can write something substantial.

What you need from a CV, cover letter, etc. depends highly on where you are applying.

- Academic CV's are typically very plain: there are several online templates that you can find for this. Look up the rules for the country you are applying to.
- Non-academic CVs are more customized, but at the end of the day, designed to highlight the qualifications that make you suitable for the job.

Random advice:

- **Get a Linked-In profile.** Look for organizations and people who work in your areas of interest.
- Do not expect people to respond to your emails: 90% of people don't.
- Professors get a lot of emails, so you have to distinguish yourself in some way. Schools and networking opportunities like this one are helpful.
- If you don't know a person personally and you're sending an email, the first or second sentence should make it clear that you have the right kind of background to work with the person you're emailing, this increases the chances that people continue reading.
- Before emailing, do some research on whether there are scholarship or partner programs between your home country/university/department and the country/university/department that you want to go to. If so, mention this in the email. Also mention potential letter writers in the email, and, if possible, try to have a letter writer that the person you're emailing might know.
- Good things to know and have (for PhD positions - postdocs are very different)
 - Positive recommendation letters and internationally recognized degrees are much much more important than papers, publications, etc., as is your background. Schools like this one help to complement your training.
 - CIMPA is also putting together a mentorship scheme that could be helpful
 - International engagement is helpful: Programmes/degrees like the AIMS Masters, IMM (International Mathematics Master), ICTP Diploma form stepping stones towards an international PhD.

AI Policy

It is very tempting to get AI to write parts of your application, but since a lot of people are doing this these days, you run the risk of getting an application that sounds the same as everyone else's. So it is important that you have your own ideas, thoughts and proposals, so that you can distinguish yourself. Once you have all of the ingredients, you can use AI as a tool for editing or spell-check or formatting: but do not use it to write the initial draft of your application.